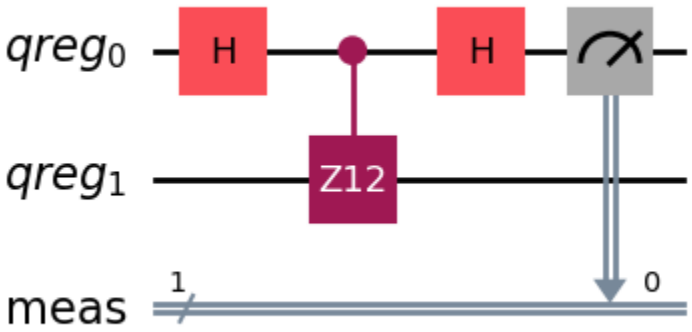
$$F = H, \qquad Z^{(12)} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad B_2 = \frac{e^{-\pi i/8}}{2}\begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix} = \frac{e^{\pi i/8}}{\sqrt{2}}\begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}$$



$$qreg_0 = |0\rangle, \qquad qreg_1 = |\psi\rangle = a|0\rangle + b|1\rangle$$

At $t = 0$:

$$\begin{aligned} |\phi_0\rangle &= |0\psi\rangle \\ &= |0\rangle \otimes (a|0\rangle + b|1\rangle) \end{aligned}$$

At $t = 1$ (after H on the first qubit):

$$\begin{aligned} |\phi_1\rangle &= (H|0\rangle) \otimes (a|0\rangle + b|1\rangle) \\ &= |+\rangle \otimes (a|0\rangle + b|1\rangle) \\ &= a|+0\rangle + b|+1\rangle \\ &= \frac{1}{\sqrt{2}}(a|00\rangle + a|10\rangle + b|01\rangle + b|11\rangle) \end{aligned}$$

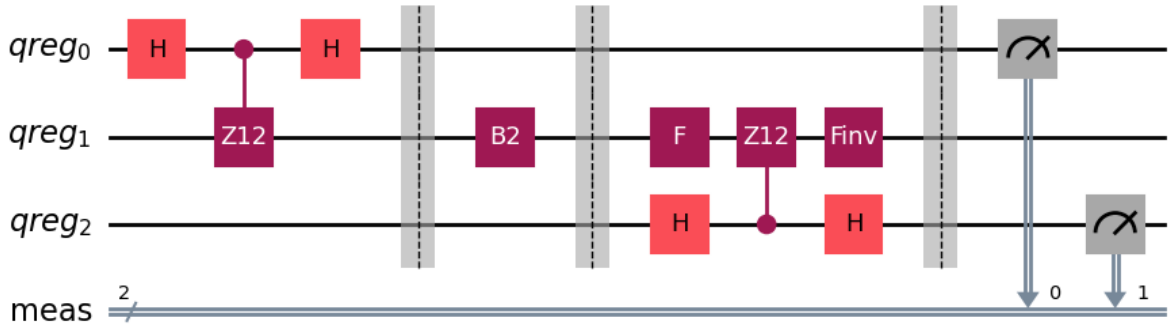
At $t = 2$ (after controlled-$Z^{(12)}$):

$$\begin{aligned} |\phi_2\rangle &= \frac{1}{\sqrt{2}}\left(a|00\rangle + a|1\rangle Z^{(12)}|0\rangle + b|01\rangle + b|1\rangle Z^{(12)}|1\rangle\right) \\ &= \frac{1}{\sqrt{2}}(a|00\rangle + a|10\rangle + b|01\rangle - b|11\rangle) \\ &= a|+0\rangle + b|-1\rangle \end{aligned}$$

At $t = 3$ (after H on the first qubit, again):

$$\begin{aligned} |\phi_3\rangle &= a(H|+\rangle)|0\rangle + b(H|-\rangle)|1\rangle \\ &= a|00\rangle + b|11\rangle \end{aligned}$$

By the Born Rule, the probability of measuring the first qubit 0 is $|a|^2$ and the first qubit 1 is $|b|^2$



$$qreg_0 = |0\rangle, \qquad qreg_1 = |\psi\rangle = a|0\rangle + b|1\rangle, \qquad qreg_2 = |0\rangle$$

At $t = 4$ (after the first dotted line barrier, before B_2):

$$|\phi_4\rangle = a|000\rangle + b|110\rangle$$

At $t = 5$ (after B_2):

$$\begin{aligned}|\phi_5\rangle &= a|0\rangle(B_2|0\rangle)|0\rangle + b|1\rangle(B_2|1\rangle)|0\rangle \\ &= \frac{e^{-\pi i/8}}{2}\big(a|0\rangle\big((1+i)|0\rangle + (1-i)|1\rangle\big)|0\rangle + b|1\rangle\big((1-i)|0\rangle + (1+i)|1\rangle\big)|0\rangle\big) \\ &= \frac{e^{-\pi i/8}}{2}(a(1+i)|000\rangle + a(1-i)|010\rangle + b(1-i)|100\rangle + b(1+i)|110\rangle)\end{aligned}$$

At $t = 6$ (after $I \otimes F \otimes H$):

$$\begin{aligned}|\phi_6\rangle &= \frac{e^{-\pi i/8}}{2}\big(a|0\rangle\big((1+i)F|0\rangle + (1-i)F|1\rangle\big)H|0\rangle + b|1\rangle\big((1-i)F|0\rangle + (1+i)F|1\rangle\big)H|0\rangle\big) \\ &= \frac{e^{-\pi i/8}}{2}\big(a|0\rangle\big((1+i)|+\rangle + (1-i)|-\rangle\big)|+\rangle + b|1\rangle\big((1-i)|+\rangle + (1+i)|-\rangle\big)|+\rangle\big) \\ &= \frac{e^{-\pi i/8}}{2\sqrt{2}}\Big(a|0\rangle\big((1+i)|+\rangle + (1-i)|-\rangle\big)(|0\rangle + |1\rangle) \\ &\qquad + b|1\rangle\big((1-i)|+\rangle + (1+i)|-\rangle\big)(|0\rangle + |1\rangle)\Big) \\ &= \frac{e^{-\pi i/8}}{2\sqrt{2}}\big(a|0\rangle\big((1+i)|+\rangle + (1-i)|-\rangle\big)|0\rangle + a|0\rangle\big((1+i)|+\rangle + (1-i)|-\rangle\big)|1\rangle \\ &\qquad + b|1\rangle\big((1-i)|+\rangle + (1+i)|-\rangle\big)|0\rangle + b|1\rangle\big((1-i)|+\rangle + (1+i)|-\rangle\big)|1\rangle\big)\end{aligned}$$

At $t = 7$ (after controlled-$Z^{(12)}$):

$$\begin{aligned}|\phi_7\rangle &= \frac{e^{-\pi i/8}}{2\sqrt{2}}\big(a|0\rangle\big((1+i)|+\rangle + (1-i)|-\rangle\big)|0\rangle + a|0\rangle Z^{(12)}\big((1+i)|+\rangle + (1-i)|-\rangle\big)|1\rangle \\ &\qquad + b|1\rangle\big((1-i)|+\rangle + (1+i)|-\rangle\big)|0\rangle + b|1\rangle Z^{(12)}\big((1-i)|+\rangle + (1+i)|-\rangle\big)|1\rangle\big) \\ &= \frac{e^{-\pi i/8}}{2\sqrt{2}}\big(a|0\rangle\big((1+i)|+\rangle + (1-i)|-\rangle\big)|0\rangle + a|0\rangle\big((1+i)|-\rangle + (1-i)|+\rangle\big)|1\rangle \\ &\qquad + b|1\rangle\big((1-i)|+\rangle + (1+i)|-\rangle\big)|0\rangle + b|1\rangle\big((1-i)|-\rangle + (1+i)|+\rangle\big)|1\rangle\big)\end{aligned}$$

At $t = 8$ (after $I \otimes F^{-1} \otimes H$):

$$
\begin{aligned}
|\phi_8\rangle = \frac{e^{-\pi i/8}}{2\sqrt{2}} & \big(a|0\rangle F^{-1}\big((1+i)|+\rangle + (1-i)|-\rangle\big)H|0\rangle \\
& + a|0\rangle F^{-1}\big((1+i)|-\rangle + (1-i)|+\rangle\big)H|1\rangle \\
& + b|1\rangle F^{-1}\big((1-i)|+\rangle + (1+i)|-\rangle\big)H|0\rangle \\
& + b|1\rangle F^{-1}\big((1-i)|-\rangle + (1+i)|+\rangle\big)H|1\rangle\big) \\
= \frac{e^{-\pi i/8}}{2\sqrt{2}} & \big(a|0\rangle\big((1+i)|0\rangle + (1-i)|1\rangle\big)|+\rangle + a|0\rangle\big((1+i)|1\rangle + (1-i)|0\rangle\big)|-\rangle \\
& + b|1\rangle\big((1-i)|0\rangle + (1+i)|1\rangle\big)|+\rangle + b|1\rangle\big((1-i)|1\rangle + (1+i)|0\rangle\big)|-\rangle\big) \\
= \frac{e^{-\pi i/8}}{4} & \Big(a|0\rangle\big((1+i)|0\rangle + (1-i)|1\rangle\big)(|0\rangle + |1\rangle) \\
& + a|0\rangle\big((1+i)|1\rangle + (1-i)|0\rangle\big)(|0\rangle - |1\rangle) \\
& + b|1\rangle\big((1-i)|0\rangle + (1+i)|1\rangle\big)(|0\rangle + |1\rangle) \\
& + b|1\rangle\big((1-i)|1\rangle + (1+i)|0\rangle\big)(|0\rangle - |1\rangle)\Big) \\
= \frac{e^{-\pi i/8}}{4} & \Big(a\big((1+i)|000\rangle + (1+i)|001\rangle + (1-i)|010\rangle + (1-i)|011\rangle\big) \\
& + a\big((1+i)|010\rangle - (1+i)|011\rangle + (1-i)|000\rangle - (1-i)|001\rangle\big) \\
& + b\big((1-i)|100\rangle + (1-i)|101\rangle + (1+i)|110\rangle + (1+i)|111\rangle\big) \\
& + b\big((1-i)|110\rangle - (1-i)|111\rangle + (1+i)|100\rangle - (1+i)|101\rangle\big)\Big) \\
= \frac{e^{-\pi i/8}}{4} & \Big(a\Big(\big((1+i)+(1-i)\big)|000\rangle + \big((1+i)-(1-i)\big)|001\rangle \\
& + \big((1-i)+(1+i)\big)|010\rangle + \big((1-i)-(1+i)\big)|011\rangle\Big) \\
& + b\Big(\big((1-i)+(1+i)\big)|100\rangle + \big((1-i)-(1+i)\big)|101\rangle \\
& + \big((1+i)+(1-i)\big)|110\rangle + \big((1+i)-(1-i)\big)|111\rangle\Big)\Big) \\
= \frac{e^{-\pi i/8}}{4} & \big(a(2|000\rangle + 2i|001\rangle + 2|010\rangle - 2i\,|011\rangle) \\
& + b(2|100\rangle - 2i|101\rangle + 2|110\rangle + 2i|111\rangle)\big) \\
= \frac{e^{-\pi i/8}}{2} & \big(a(|000\rangle + i|001\rangle + |010\rangle - i\,|011\rangle) + b(|100\rangle - i|101\rangle + |110\rangle + i|111\rangle)\big)
\end{aligned}
$$

Thus, probability of measuring 0 on the third qubit is:

$$p_3(0) = \left|\frac{a}{2}\right|^2 + \left|\frac{a}{2}\right|^2 + \left|\frac{b}{2}\right|^2 + \left|\frac{b}{2}\right|^2 = \frac{|a|^2 + |b|^2}{2} = \frac{1}{2}$$

If B_2 was omitted, at $t = 6$:

$$
\begin{aligned}
|\phi_6'\rangle &= a|0\rangle F|0\rangle H|0\rangle + b|1\rangle F|1\rangle H|0\rangle \\
&= a|0++\rangle + b|1-+\rangle \\
&= \frac{1}{\sqrt{2}}(a|0+\rangle + b|1-\rangle)(|0\rangle + |1\rangle) \\
&= \frac{1}{\sqrt{2}}(a|0+\rangle + b|1-\rangle)|0\rangle + \frac{1}{\sqrt{2}}(a|0+\rangle + b|1-\rangle)|1\rangle
\end{aligned}
$$

At $t = 7$:

$$|\phi_7'\rangle = \frac{1}{\sqrt{2}}(a|0+\rangle + b|1-\rangle)|0\rangle + \frac{1}{\sqrt{2}}\left(a|0\rangle Z^{(12)}|+\rangle + b|1\rangle Z^{(12)}|-\rangle\right)|1\rangle$$
$$= \frac{1}{\sqrt{2}}\left((a|0+\rangle + b|1-\rangle)|0\rangle + (a|0-\rangle + b|1+\rangle)|1\rangle\right)$$

At $t = 8$:

$$|\phi_8'\rangle = \frac{1}{\sqrt{2}}\left((a|0\rangle F|+\rangle + b|1\rangle F|-\rangle)H|0\rangle + (a|0\rangle F|-\rangle + b|1\rangle F|+\rangle)H|1\rangle\right)$$
$$= \frac{1}{\sqrt{2}}\left((a|00\rangle + b|11\rangle)|+\rangle + (a|01\rangle + b|10\rangle)|-\rangle\right)$$
$$= \frac{1}{2}(a|000\rangle + a|001\rangle + b|110\rangle + b|111\rangle + a|010\rangle - a|011\rangle + b|100\rangle - b|101\rangle)$$

Applying Born rule to compute the probability of measuring 0 on the third qubit is:

$$p_3'(0) = \left|\frac{a}{2}\right|^2 + \left|\frac{b}{2}\right|^2 + \left|\frac{a}{2}\right|^2 + \left|\frac{b}{2}\right|^2 = \frac{1}{2}$$

If the first qubit is ignored (i.e., 2-qubit circuit with just $qreg_1$ and $qreg_2$)

When B_2 is included, we have $|\phi_8\rangle$ as follows:

$$|\phi_8\rangle = \frac{e^{-\pi i/8}}{2}\left(a(|00\rangle + i|01\rangle + |10\rangle - i\,|11\rangle) + b(|00\rangle - i|01\rangle + |10\rangle + i|11\rangle)\right)$$
$$= \frac{e^{-\pi i/8}}{2}\left((a+b)|00\rangle + i(a-b)|01\rangle + (a+b)|10\rangle + i(-a+b)|11\rangle\right)$$

Thus, $p_3(0)$ becomes:

$$p_3(0) = \left|\frac{a+b}{2}\right|^2 + \left|\frac{a+b}{2}\right|^2 = \left|\frac{a+b}{\sqrt{2}}\right|^2$$

When B_2 is omitted, we have $|\phi_8'\rangle$ as follows:

$$|\phi_8'\rangle = \frac{1}{2}(a|00\rangle + a|01\rangle + b|10\rangle + b|11\rangle + a|10\rangle - a|11\rangle + b|00\rangle - b|01\rangle)$$
$$= \frac{1}{2}\left((a+b)|00\rangle + (a-b)|01\rangle + (a+b)|10\rangle + (-a+b)|11\rangle\right)$$

Thus, $p_3'(0)$ becomes:

$$p_3'(0) = \left|\frac{a+b}{2}\right|^2 + \left|\frac{a+b}{2}\right|^2 = \left|\frac{a+b}{\sqrt{2}}\right|^2$$